

Akshat Rajani

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021-Present	B.Tech	Indian Institute of Technology Kanpur	9.36/10
2020-21	Class XII (CBSE)	Kautilya Sr. Sec. School, Kota	92.4%
2018-19	Class X (CBSE)	St. Joseph Public School, Kota	97.2%

ACHIEVEMENTS & SCHOLARSHIPS

- Conferred with the **Academic Excellence Award** by IIT Kanpur for outstanding academic performance
- Secured **All India Rank 186** in **JEE Advanced 2021**, held by IIT Kharagpur, among 1.4 Lakh candidates
- Secured **All India Rank 220** in **JEE Mains 2021**, conducted by NTA among 10 Lakh potential candidates
- Qualified for **KVPY Fellowship Award 2020** by securing **AIR 189** in SX-stream among 40,000 participants
- Qualified for **Indian National Mathematical Olympiad (INMO)**, 2020 conducted by HBCSE, Mumbai
- Awarded **NTS Scholarship** for securing top 2000 position in **National Talent Search Examination**, 2019
- Received **Certificate of Merit** from CBSE for scoring **100/100** in Mathematics in Class X Board Examination

KEY PROJECTS

- Campus Pay | Course Project, CS253, Prof. Indranil Saha, IIT Kanpur Jan'23 - Apr'23
- Collaborated in a team of 10 members to design and develop a **Full-Stack Web Application** from scratch to **digitalize dues management** system between shopkeepers and campus community at IIT Kanpur
 - Used software utilities such as **ReactJS** for front-end, **Django** for back-end, **PostgreSQL** for database management during implementation, and developed proficiency in **GitHub**, ensuring smooth project management
 - Maintained **Requirement Specifications**, **Design**, and **Beta-Testing Document** throughout the project
- Mathematical Trading Strategies | Stamatics, IIT Kanpur May'23 - Jul'23
- Designed and implemented an effective **Python trading strategy** using a combination of **Keltner Channel** and **Hammer Candlestick Patterns** yielding a **sharpe ratio** of **1.04** and **cumulative returns** of **1.58**
 - Generated buy/sell signals for **NASDAQ** by optimizing indicator parameters and cross-correlation with **NSE**
 - Evaluated various technical indicators like **Bollinger Bands** and **MACD**, for optimal **performance metrics**
 - Identified different patterns in prices like **inverse cup and handle**, and **rounding bottom** from Stock Charts
- Uncovering Mask of XORRO|ML Mini-Project, CS771, Prof. Purushottam Kar, IIT Kanpur Feb'23 - Feb'23
- Cracked a **Physically Unclonable Function** made of **2⁸** XOR Ring Oscillators using multiple linear classifiers
 - Analyzed training time and accuracy of LinearSVC and LogisticRegression and the effect of **hyperparameters**
- Playing with the Melbot | ML Mini-Project, CS771, Prof. Purushottam Kar, IIT Kanpur Mar'23 - Mar'23
- Pickled an ML model that accurately predicts any word from a dictionary containing more than **5000** words
 - Implemented **Decision Tree algorithm** using strategies such as **entropy reduction** and **lookahead** to optimize model parameters such as **training time**, **win rate**, **average query count** and the **on-disk size**
- CSE-Bubble | Course Project, CS220, Prof. Urbi Chatterjee Jan'23 - Apr'23
- Implemented **Verilog Hardware Description** of a **32-bit Processor** featuring ISA, ALU and memory unit
 - Designed the instruction layout for **R-, I- and J-type** instructions and their respective encoding methodologies
 - Successfully built a **single-cycle instruction execution unit**, reinforcing the comprehensive processor design

RELEVANT SKILLS

Languages: C | C++ | Python | Bash | Html | CSS | Javascript | Verilog | MIPS | L^AT_EX | Arduino

Softwares/Libraries: Numpy | Pandas | Scikit Learn | Matplotlib | ReactJS | GitHub | Autodesk Fusion 360

RELEVANT COURSEWORK

(A*)-Outstanding Performance (among top 5% of class)

Introduction to Probability	Data Structures and Algorithms	Introduction to Machine Learning (A*)
Fundamental of Computing	Mathematics for Computer Science	Software Development And Operations
Introduction to Electronics	Real Analysis and Calculus	Linear Algebra and Differential Equation

POSITIONS OF RESPONSIBILITY

- Secretary, Career Development Wing | Academic and Career Council, IIT Kanpur Jun'22 - Apr'23
- Organised and managed talks and competitions under Career Connect, the flagship event, with council members
 - Revised the databases of speakers and alums and designed posters using Canva for session announcements